Cardiology_Database_Management

Database Design & SQL Query Project

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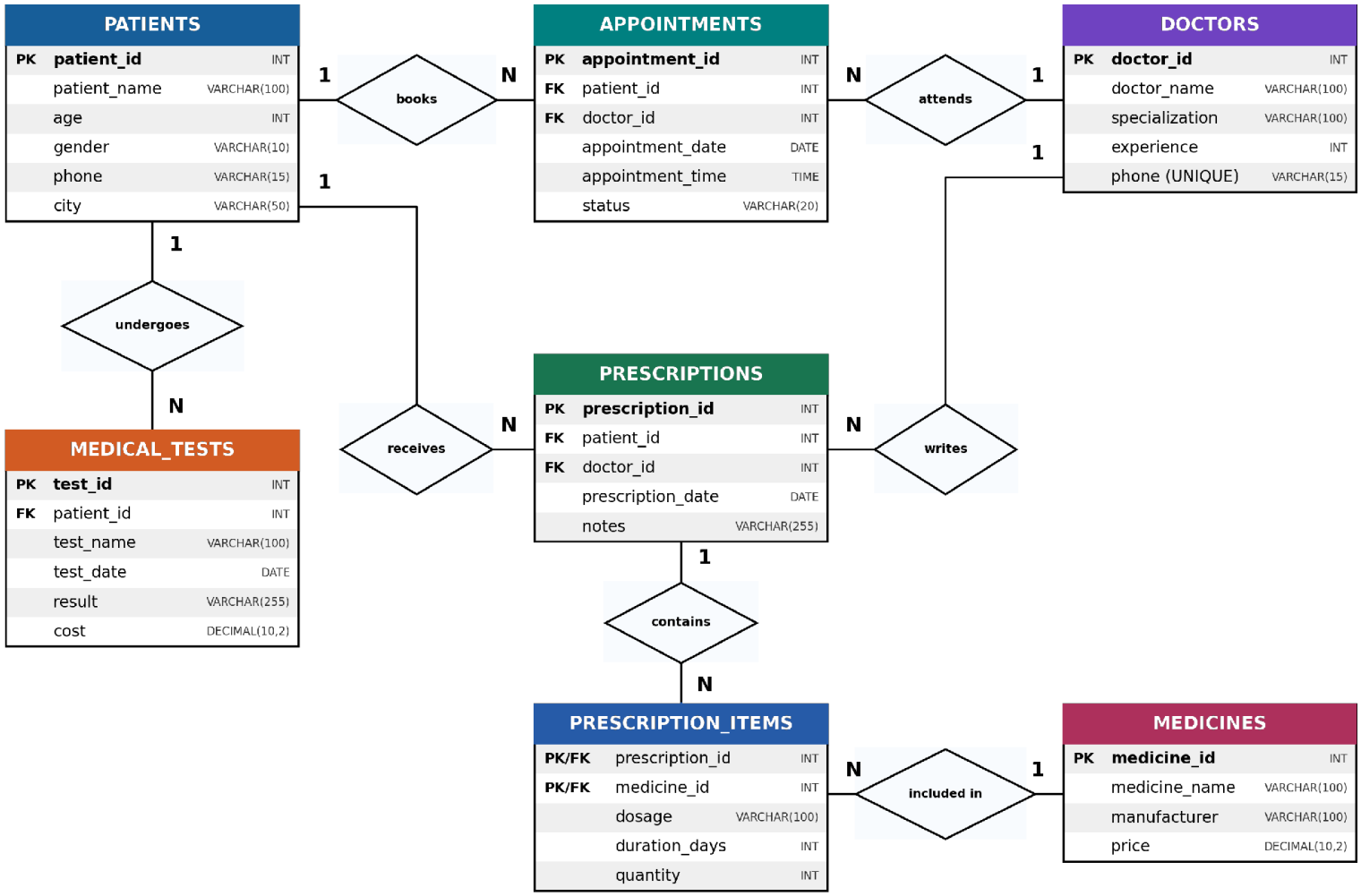
Theme

A cardiology clinic wants to maintain patient and doctor details, schedule appointments, record diagnostic tests, and keep prescriptions with the medicines prescribed, so that patient care, test costs and doctor workload can be analysed from one database (cardiology_db, MySQL InnoDB).

Modules

- **Patients** – patient name, age, gender, phone and city
- **Doctors** – doctor name, specialization, experience and phone
- **Appointments** – date, time and status of each patient–doctor visit
- **Medical Tests** – test name, date, result and cost for each patient
- **Prescriptions** – prescription date and notes written by a doctor for a patient
- **Medicines** – medicine catalogue with manufacturer and price
- **Prescription Items** – medicines, dosage, duration and quantity in each prescription

ER Diagram



ER Diagram – Cardiology Database Management (PK = Primary Key, FK = Foreign Key)

Relationships

- **Patients → Appointments:** One patient can have many appointments (1 : N).
- **Doctors → Appointments:** One doctor can have many appointments (1 : N).
- **Patients → Medical Tests:** One patient can undergo many tests (1 : N).
- **Patients → Prescriptions:** One patient can receive many prescriptions (1 : N).
- **Doctors → Prescriptions:** One doctor can write many prescriptions (1 : N).
- **Prescriptions → Prescription Items:** One prescription can contain many items (1 : N).
- **Medicines → Prescription Items:** One medicine can appear in many items (1 : N).

Prescription Items is the bridge table that makes Prescriptions and Medicines a many-to-many relationship. Its primary key is the pair (prescription_id, medicine_id).

Table Structure

PATIENTS

Column	Data Type	Key
patient_id	INT	Primary Key
patient_name	VARCHAR(100)	NOT NULL
age	INT	
gender	VARCHAR(10)	
phone	VARCHAR(15)	
city	VARCHAR(50)	

DOCTORS

Column	Data Type	Key
doctor_id	INT	Primary Key
doctor_name	VARCHAR(100)	NOT NULL
specialization	VARCHAR(100)	
experience	INT	Years of experience
phone	VARCHAR(15)	Unique

APPOINTMENTS

Column	Data Type	Key
appointment_id	INT	Primary Key
patient_id	INT	Foreign Key → patients
doctor_id	INT	Foreign Key → doctors
appointment_date	DATE	NOT NULL
appointment_time	TIME	
status	VARCHAR(20)	Default 'Scheduled'

MEDICAL_TESTS

Column	Data Type	Key
test_id	INT	Primary Key
patient_id	INT	Foreign Key → patients
test_name	VARCHAR(100)	NOT NULL
test_date	DATE	
result	VARCHAR(255)	
cost	DECIMAL(10,2)	

PRESCRIPTIONS

Column	Data Type	Key
prescription_id	INT	Primary Key
patient_id	INT	Foreign Key → patients
doctor_id	INT	Foreign Key → doctors

Column	Data Type	Key
prescription_date	DATE	NOT NULL
notes	VARCHAR(255)	

MEDICINES

Column	Data Type	Key
medicine_id	INT	Primary Key
medicine_name	VARCHAR(100)	NOT NULL
manufacturer	VARCHAR(100)	
price	DECIMAL(10,2)	

PRESCRIPTION_ITEMS

Column	Data Type	Key
prescription_id	INT	Primary Key (part 1), Foreign Key → prescriptions
medicine_id	INT	Primary Key (part 2), Foreign Key → medicines
dosage	VARCHAR(100)	
duration_days	INT	
quantity	INT	

Sample Data

PATIENTS

patient_id	patient_name	age	gender	phone	city
1	Rahul Kumar	45	Male	9876501001	Srikakulam
2	Priya Sharma	38	Female	9876501002	Visakhapatnam
3	Arjun Reddy	55	Male	9876501003	Vijayawada
4	Sneha Rao	42	Female	9876501004	Hyderabad
5	Kiran Kumar	60	Male	9876501005	Srikakulam

DOCTORS

doctor_id	doctor_name	specialization	experience	phone
1	Dr. Suresh Reddy	Cardiologist	15	9888001001
2	Dr. Anitha Rao	Interventional Cardiologist	12	9888001002
3	Dr. Rajesh Kumar	Cardiac Surgeon	18	9888001003
4	Dr. Lakshmi Devi	Cardiologist	10	9888001004

APPOINTMENTS

appointment_id	patient_id	doctor_id	appointment_date	appointment_time	status
1	1	1	2026-11-02	09:30:00	Scheduled
2	2	2	2026-11-02	10:00:00	Scheduled
3	3	3	2026-11-03	11:00:00	Scheduled
4	4	1	2026-11-04	12:00:00	Scheduled
5	5	4	2026-11-05	09:00:00	Scheduled

MEDICAL_TESTS

test_id	patient_id	test_name	test_date	result	cost
1	1	ECG	2026-10-20	Normal rhythm	500.00
2	2	Echocardiogram	2026-10-21	Requires doctor review	2500.00
3	3	Lipid Profile	2026-10-22	Cholesterol elevated	800.00
4	4	Stress Test	2026-10-23	Requires doctor review	1800.00
5	5	ECG	2026-10-24	Requires doctor review	500.00

PRESCRIPTIONS

prescription_id	patient_id	doctor_id	prescription_date	notes
1	1	1	2026-10-20	Sample prescription; follow the treating doctor advice
2	2	2	2026-10-21	Sample prescription; follow the treating doctor advice
3	3	3	2026-10-22	Sample prescription; follow the treating doctor advice
4	4	1	2026-10-23	Sample prescription; follow the treating doctor advice
5	5	4	2026-10-24	Sample prescription; follow the treating doctor advice

MEDICINES

medicine_id	medicine_name	manufacturer	price
1	Aspirin	Sample Pharma A	50.00
2	Atorvastatin	Sample Pharma B	120.00
3	Metoprolol	Sample Pharma C	90.00
4	Amlodipine	Sample Pharma D	75.00
5	Clopidogrel	Sample Pharma E	110.00

PRESCRIPTION_ITEMS

prescription_id	medicine_id	dosage	duration_days	quantity
1	1	As entered by the treating doctor	5	5
1	2	As entered by the treating doctor	30	30
2	3	As entered by the treating doctor	7	14
3	4	As entered by the treating doctor	10	10
4	5	As entered by the treating doctor	15	15
5	2	As entered by the treating doctor	30	30

All dosage and result values are placeholder sample data for learning purposes.

Queries

1. Display All Patients

```
SELECT * FROM patients;
```

Output

patient_id	patient_name	age	gender	phone	city
1	Rahul Kumar	45	Male	9876501001	Srikakulam
2	Priya Sharma	38	Female	9876501002	Visakhapatnam
3	Arjun Reddy	55	Male	9876501003	Vijayawada
4	Sneha Rao	42	Female	9876501004	Hyderabad
5	Kiran Kumar	60	Male	9876501005	Srikakulam

2. Display All Doctors

```
SELECT * FROM doctors;
```

Output

doctor_id	doctor_name	specialization	experience	phone
1	Dr. Suresh Reddy	Cardiologist	15	9888001001
2	Dr. Anitha Rao	Interventional Cardiologist	12	9888001002
3	Dr. Rajesh Kumar	Cardiac Surgeon	18	9888001003
4	Dr. Lakshmi Devi	Cardiologist	10	9888001004

3. Display Appointments with Patient and Doctor Names

```
SELECT a.appointment_id,
       p.patient_name,
       d.doctor_name,
       a.appointment_date,
       a.appointment_time,
       a.status
FROM appointments AS a
JOIN patients AS p ON a.patient_id = p.patient_id
JOIN doctors AS d ON a.doctor_id = d.doctor_id
ORDER BY a.appointment_date, a.appointment_time;
```

Output

appointment_id	patient_name	doctor_name	appointment_date	appointment_time	status
1	Rahul Kumar	Dr. Suresh Reddy	2026-11-02	09:30:00	Scheduled
2	Priya Sharma	Dr. Anitha Rao	2026-11-02	10:00:00	Scheduled
3	Arjun Reddy	Dr. Rajesh Kumar	2026-11-03	11:00:00	Scheduled
4	Sneha Rao	Dr. Suresh Reddy	2026-11-04	12:00:00	Scheduled
5	Kiran Kumar	Dr. Lakshmi Devi	2026-11-05	09:00:00	Scheduled

4. Display Medical Test Details with Patient Names

```
SELECT p.patient_name,
       mt.test_name,
       mt.test_date,
       mt.result,
       mt.cost
FROM medical_tests AS mt
JOIN patients AS p ON mt.patient_id = p.patient_id
ORDER BY mt.test_date;
```

Output

patient_name	test_name	test_date	result	cost
Rahul Kumar	ECG	2026-10-20	Normal rhythm	500.00
Priya Sharma	Echocardiogram	2026-10-21	Requires doctor review	2500.00
Arjun Reddy	Lipid Profile	2026-10-22	Cholesterol elevated	800.00
Sneha Rao	Stress Test	2026-10-23	Requires doctor review	1800.00
Kiran Kumar	ECG	2026-10-24	Requires doctor review	500.00

5. Display Prescription and Medicine Details

```
SELECT p.patient_name,
       d.doctor_name,
       pr.prescription_date,
       m.medicine_name,
       pi.dosage,
       pi.duration_days,
       pi.quantity,
       m.price
FROM prescription_items AS pi
JOIN prescriptions AS pr ON pi.prescription_id = pr.prescription_id
JOIN patients AS p ON pr.patient_id = p.patient_id
JOIN doctors AS d ON pr.doctor_id = d.doctor_id
JOIN medicines AS m ON pi.medicine_id = m.medicine_id
ORDER BY pr.prescription_id, m.medicine_id;
```

Output

patient_name	doctor_name	prescription_date	medicine_name	dosage	duration_days	quantity	price
Rahul Kumar	Dr. Suresh Reddy	2026-10-20	Aspirin	As entered by the treating doctor	5	5	50.00
Rahul Kumar	Dr. Suresh Reddy	2026-10-20	Atorvastatin	As entered by the treating doctor	30	30	120.00
Priya Sharma	Dr. Anitha Rao	2026-10-21	Metoprolol	As entered by the treating doctor	7	14	90.00
Arjun Reddy	Dr. Rajesh Kumar	2026-10-22	Amlodipine	As entered by the treating doctor	10	10	75.00
Sneha Rao	Dr. Suresh Reddy	2026-10-23	Clopidogrel	As entered by the treating doctor	15	15	110.00
Kiran Kumar	Dr. Lakshmi Devi	2026-10-24	Atorvastatin	As entered by the treating doctor	30	30	120.00

6. Total Medical Test Cost for Each Patient

```
SELECT p.patient_name,
       SUM(mt.cost) AS total_test_cost
FROM patients AS p
JOIN medical_tests AS mt ON p.patient_id = mt.patient_id
GROUP BY p.patient_id, p.patient_name;
```

Output

patient_name	total_test_cost
Rahul Kumar	500.00
Priya Sharma	2500.00
Arjun Reddy	800.00
Sneha Rao	1800.00
Kiran Kumar	500.00

7. Display Patients Older Than 40 Years

```
SELECT *
FROM patients
WHERE age > 40;
```

Output

patient_id	patient_name	age	gender	phone	city
1	Rahul Kumar	45	Male	9876501001	Srikakulam
3	Arjun Reddy	55	Male	9876501003	Vijayawada
4	Sneha Rao	42	Female	9876501004	Hyderabad
5	Kiran Kumar	60	Male	9876501005	Srikakulam

8. Count the Total Number of Patients

```
SELECT COUNT(*) AS total_patients
FROM patients;
```

Output

total_patients
5

9. Doctors with More Than 10 Years of Experience

```
SELECT doctor_name, specialization, experience
FROM doctors
WHERE experience > 10;
```

Output

doctor_name	specialization	experience
Dr. Suresh Reddy	Cardiologist	15
Dr. Anitha Rao	Interventional Cardiologist	12
Dr. Rajesh Kumar	Cardiac Surgeon	18

10. Count Appointments for Each Doctor

```
SELECT d.doctor_name,
       COUNT(a.appointment_id) AS total_appointments
FROM doctors AS d
LEFT JOIN appointments AS a ON d.doctor_id = a.doctor_id
GROUP BY d.doctor_id, d.doctor_name;
```

Output

doctor_name	total_appointments
Dr. Suresh Reddy	2
Dr. Anitha Rao	1
Dr. Rajesh Kumar	1
Dr. Lakshmi Devi	1

11. Display Scheduled Appointments

```
SELECT *
FROM appointments
WHERE status = 'Scheduled';
```

Output

appointment_id	patient_id	doctor_id	appointment_date	appointment_time	status
1	1	1	2026-11-02	09:30:00	Scheduled

appointment_id	patient_id	doctor_id	appointment_date	appointment_time	status
2	2	2	2026-11-02	10:00:00	Scheduled
3	3	3	2026-11-03	11:00:00	Scheduled
4	4	1	2026-11-04	12:00:00	Scheduled
5	5	4	2026-11-05	09:00:00	Scheduled

12. Display Medicines Costing More Than ₹80

```
SELECT medicine_name, manufacturer, price
FROM medicines
WHERE price > 80;
```

Output

medicine_name	manufacturer	price
Atorvastatin	Sample Pharma B	120.00
Metoprolol	Sample Pharma C	90.00
Clopidogrel	Sample Pharma E	110.00

13. Display Each Patient's Medical Tests and Costs

```
SELECT p.patient_name,
       mt.test_name,
       mt.cost
FROM patients AS p
JOIN medical_tests AS mt ON p.patient_id = mt.patient_id
ORDER BY p.patient_name;
```

Output

patient_name	test_name	cost
Arjun Reddy	Lipid Profile	800.00
Kiran Kumar	ECG	500.00
Priya Sharma	Echocardiogram	2500.00
Rahul Kumar	ECG	500.00
Sneha Rao	Stress Test	1800.00

14. Display Doctors in Alphabetical Order

```
SELECT *
FROM doctors
ORDER BY doctor_name ASC;
```

Output

doctor_id	doctor_name	specialization	experience	phone
2	Dr. Anitha Rao	Interventional Cardiologist	12	9888001002
4	Dr. Lakshmi Devi	Cardiologist	10	9888001004
3	Dr. Rajesh Kumar	Cardiac Surgeon	18	9888001003
1	Dr. Suresh Reddy	Cardiologist	15	9888001001

15. Display All Tables

```
SHOW TABLES;
```

Output

Tables_in_cardiology_db
appointments

Tables_in_cardiology_db
doctors
medical_tests
medicines
patients
prescription_items
prescriptions

PROJECT SUMMARY

The Cardiology Database Management project organizes the main activities of a cardiology clinic in a MySQL InnoDB database named cardiology_db.

• **Database design:** Seven tables are used: Patients, Doctors, Appointments, Medical_Tests, Prescriptions, Medicines, and Prescription_Items.

• **Relationships:** Patients and doctors connect to appointments; patients connect to medical tests and prescriptions; doctors connect to prescriptions. Prescription_Items links prescriptions and medicines.

• **Keys and integrity:** Primary keys identify records, while foreign keys connect related tables. Prescription_Items uses a composite primary key made from prescription_id and medicine_id.

• **Sample data:** The document includes example records for patients, doctors, appointments, tests, prescriptions, medicines, and prescription items. Values are for learning purposes.

• **SQL practice:** The examples demonstrate selecting records, joining tables, filtering with WHERE, sorting with ORDER BY, grouping and aggregation, counting records, and listing database tables.

• **Purpose:** Together, the schema and queries demonstrate how clinic information can be stored consistently and queried to review appointments, test costs, prescriptions, medicines, and doctor workloads.

Note: Sample patient details, test results, and prescription values are illustrative and should not be treated as real clinical advice.